Perimeter, Area and Composite Shapes

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS perimeter *(paenga)* · area *(horahanga)* · composite shape · square units *(waeine pūrua)*

A Perimeter and area ●○○

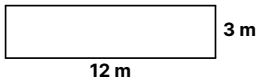
⚠ **Watch out:** Perimeter is the distance around the outside, in cm or m. Area is the space inside, in cm² or m².

1. Find the perimeter and the area of each rectangle.

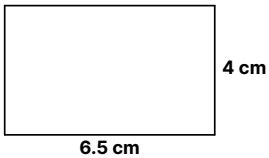
a)



b)



c)



B From perimeter to area, and back ●●○

💡 **Remember:** Find the missing side first. A square has four equal sides, so divide its perimeter by 4.

2. Answer each question.

a) A square has a perimeter of 36 cm. Find its area.

b) A square has an area of 49 cm². Find its perimeter.

c) A rectangle has a perimeter of 30 cm and a length of 9 cm. Find its area.

d) A rectangle has an area of 48 m² and a width of 6 m. Find its perimeter.

e) A square has a perimeter of 10 m. Find its area.

f) A rectangle has a perimeter of 22 m and a length of 7 m. Find its area.

C Missing lengths ●●○

💡 **Remember:** Work backwards. If area = length × width, then width = area ÷ length.

3. Find the missing length.

a) A rectangle has an area of 54 cm² and a length of 9 cm. Find its width.

b) A rectangle has a perimeter of 26 cm and a length of 8 cm. Find its width.

c) A triangle has an area of 24 cm² and a base of 8 cm. Find its height.

d) A parallelogram has an area of 60 m² and a base of 12 m. Find its height.

e) A square has a perimeter of 34 cm. Find the length of its sides.

f) A rectangle has an area of 31.5 m² and a length of 7 m. Find its width.

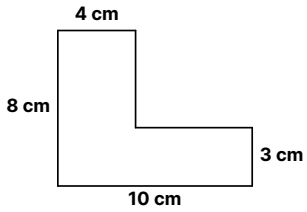
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D Composite shapes ●●●

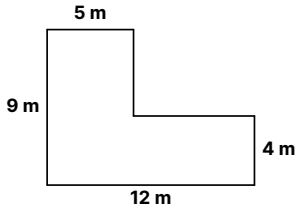
> **Remember:** Split the shape into rectangles and triangles, find each area, then add them. For a shape with a piece cut out, subtract instead.

4. All the corners are right angles. Find the perimeter and the area of each shape.

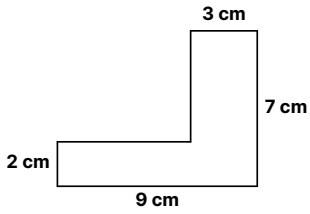
a)



b)

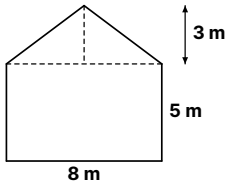


c)

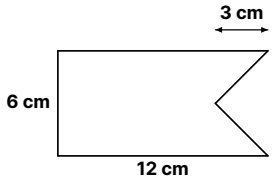


5. Find the area of each shape.

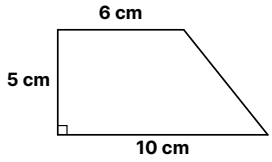
a)



b)



c)



E Problems ●●●

6. A rectangular lawn is 12 m long and 8 m wide. A square flower bed with sides of 3 m is cut out of it. Find the area of grass that is left.

7. A wall is 6 m long and 2.5 m high. It has a window that is 1.5 m wide and 1 m high. Find the area of the wall to be painted.

8. A rectangular garden is 9 m long and 6 m wide. A fence goes all the way around it, except for a gate that is 2 m wide. How long is the fence?

★ Extension

9. A rectangle has a perimeter of 24 cm. Its length is twice its width. Find its area.

10. A square and a rectangle both have an area of 36 cm². The rectangle is 9 cm long. Which has the longer perimeter? Show your working.

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Perimeter, Area and Composite Shapes

Name _____ Date _____ No calculator Version 2

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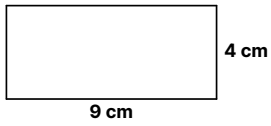
KEY WORDS perimeter *(paenga)* · area *(horahanga)* · composite shape · square units *(waeine pūrua)*

A Perimeter and area ●○○

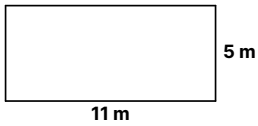
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1. Find the perimeter and the area of each rectangle.

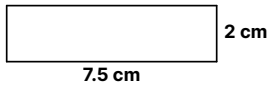
a)



b)



c)



B From perimeter to area, and back ●●○

💡 **Remember:** Find the missing side first. A square has four equal sides, so divide its perimeter by 4.

2. Answer each question.

a) A square has a perimeter of 28 cm. Find its area.

b) A square has an area of 81 cm². Find its perimeter.

c) A rectangle has a perimeter of 34 cm and a length of 10 cm. Find its area.

d) A rectangle has an area of 56 m² and a width of 7 m. Find its perimeter.

e) A square has a perimeter of 14 m. Find its area.

f) A rectangle has a perimeter of 26 m and a length of 8 m. Find its area.

C Missing lengths ●●○

💡 **Remember:** Work backwards. If area = length × width, then width = area ÷ length.

3. Find the missing length.

a) A rectangle has an area of 63 cm² and a length of 9 cm. Find its width.

b) A rectangle has a perimeter of 32 cm and a length of 9 cm. Find its width.

c) A triangle has an area of 30 cm² and a base of 10 cm. Find its height.

d) A parallelogram has an area of 72 m² and a base of 12 m. Find its height.

e) A square has a perimeter of 46 cm. Find the length of its sides.

f) A rectangle has an area of 38.5 m² and a length of 7 m. Find its width.

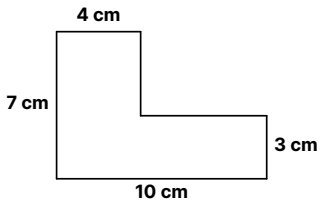
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D Composite shapes ●●●

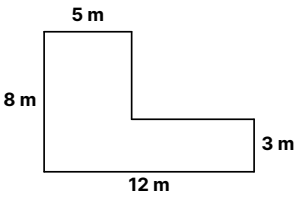
> **Remember:** Split the shape into rectangles and triangles, find each area, then add them. For a shape with a piece cut out, subtract instead.

4. All the corners are right angles. Find the perimeter and the area of each shape.

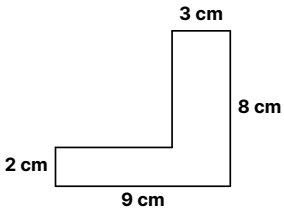
a)



b)

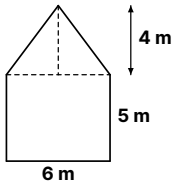


c)

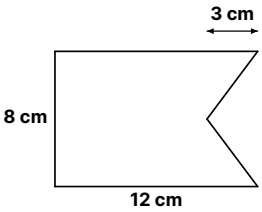


5. Find the area of each shape.

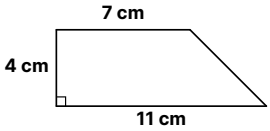
a)



b)



c)



E Problems ●●●

6. A rectangular lawn is 14 m long and 9 m wide. A square flower bed with sides of 4 m is cut out of it. Find the area of grass that is left.

7. A wall is 7 m long and 2.5 m high. It has a window that is 2 m wide and 1 m high. Find the area of the wall to be painted.

8. A rectangular garden is 10 m long and 7 m wide. A fence goes all the way around it, except for a gate that is 2 m wide. How long is the fence?

★ Extension

9. A rectangle has a perimeter of 36 cm. Its length is twice its width. Find its area.

10. A square and a rectangle both have an area of 64 cm². The rectangle is 16 cm long. Which has the longer perimeter? Show your working.

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